

AI WeatherWise API Project

Python + Flask Backend

1. Project Files

```
AI-WeatherWise/  
■■■ app.py  
■■■ requirements.txt  
■■■ .env  
■■■ README.md
```

2. app.py

```
from flask import Flask, request, jsonify  
from dotenv import load_dotenv  
import requests  
import os  
  
load_dotenv()  
  
app = Flask(__name__)  
  
WEATHER_API_KEY = os.getenv("WEATHER_API_KEY")  
  
@app.route("/")  
def home():  
    return "AI WeatherWise API is running."  
  
@app.route("/api/weather", methods=["GET"])  
def get_weather():  
    city = request.args.get("city")  
  
    if not city:  
        return jsonify({"error": "City name is required"}), 400  
  
    url = "https://api.openweathermap.org/data/2.5/weather"  
  
    params = {  
        "q": city,  
        "appid": WEATHER_API_KEY,  
        "units": "metric"  
    }  
  
    response = requests.get(url, params=params)  
  
    if response.status_code != 200:  
        return jsonify({"error": "Weather information not found"}), 404  
  
    data = response.json()  
  
    result = {  
        "city": data["name"],  
        "temperature": data["main"]["temp"],  
        "feels_like": data["main"]["feels_like"],  
        "humidity": data["main"]["humidity"],  
        "weather": data["weather"][0]["description"],  
        "wind_speed": data["wind"]["speed"]  
    }  
  
    return jsonify(result)  
  
if __name__ == "__main__":  
    app.run(debug=True)
```

3. requirements.txt

```
flask
```

```
requests
python-dotenv
```

4. .env

```
WEATHER_API_KEY=your_openweathermap_api_key
```

5. Test API

```
pip install -r requirements.txt
python app.py
```

```
http://127.0.0.1:5000/api/weather?city=Madurai
```

6. Example Response

```
{
  "city": "Madurai",
  "temperature": 30.5,
  "feels_like": 33.2,
  "humidity": 65,
  "weather": "scattered clouds",
  "wind_speed": 3.5
}
```

Important Note

Keep the weather API key in the .env file rather than directly inside app.py.